

Forecasting Australian Monthly Retail Turnover

MATH1318 Time Series Analysis | Final Project Presentation

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Dataset: Australian Bureau of Statistics, Retail Trade Australia

Research Question and Dataset

Research question

Which time series model provides the most accurate forecasts for Australian monthly retail turnover for the next 10 months?

Forecast horizon: Jul 2025 to Apr 2026

- Public source: Australian Bureau of Statistics Retail Trade, Australia workbook
- Series used: Total (Industry), Seasonally Adjusted turnover
- Monthly data from Apr 1982 to Jun 2025
- 519 observations; units are \$ million
- Seasonally adjusted series reduces calendar and recurring monthly effects

Visual and Descriptive Analysis

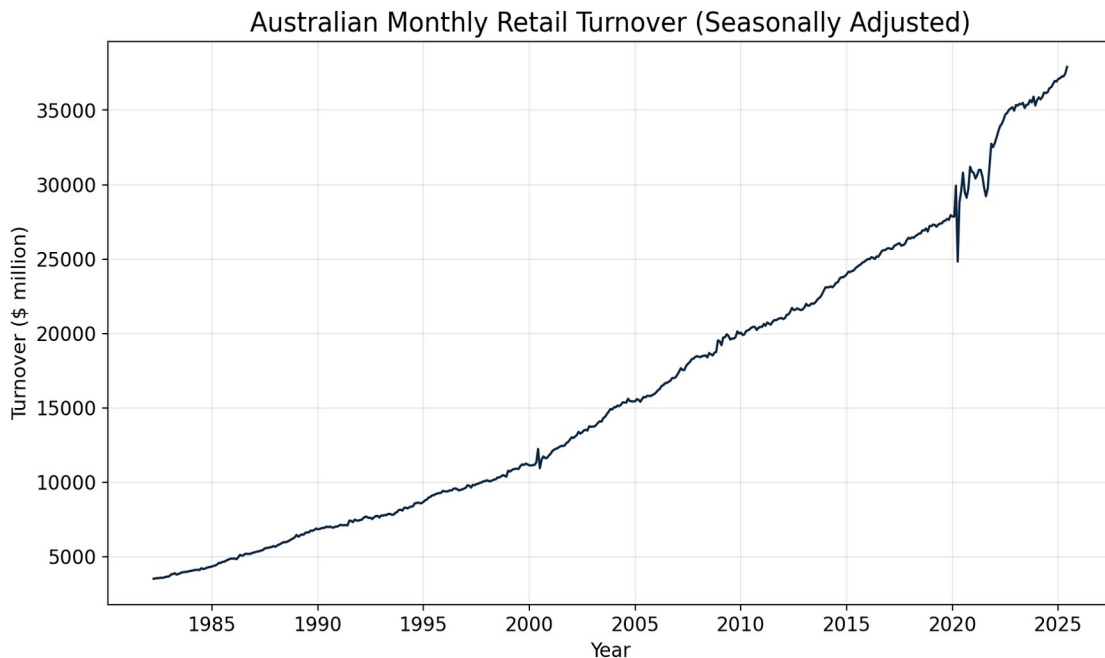


Figure 1. Seasonally adjusted total retail turnover, Australia.

- Strong long-run upward trend in retail turnover
- Series is non-stationary in mean
- Large disruptions around 2020–2021 indicate temporary shocks
- Seasonality is less visible because the selected series is seasonally adjusted
- Trend handling and differencing are required before ARIMA modelling

Model Specification

1 Transform

Log transform used to stabilise the scale of the series.

2 Difference

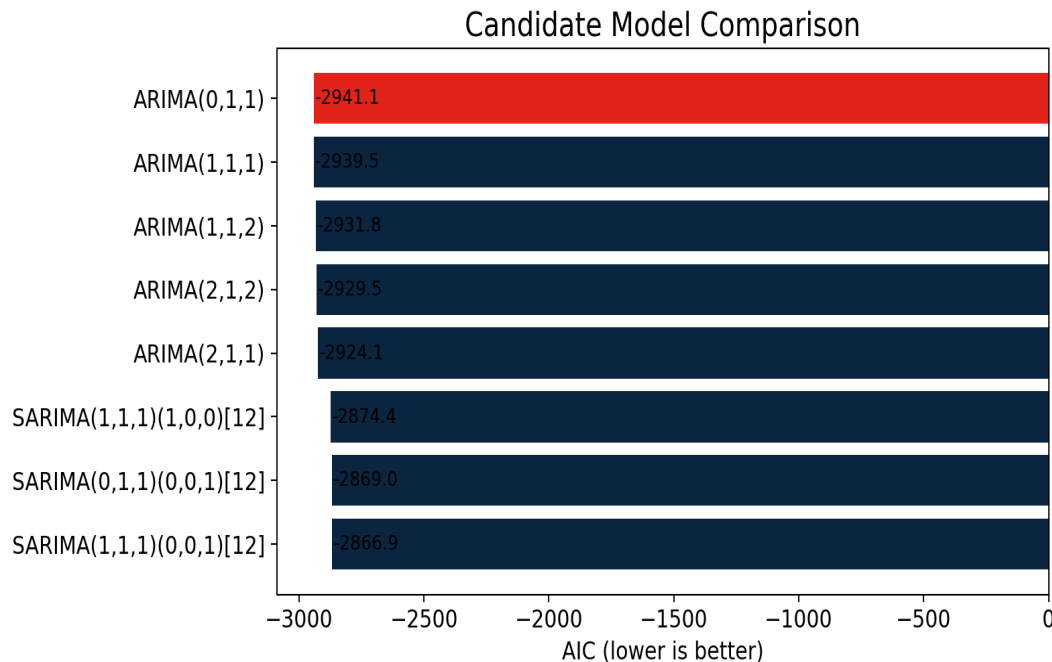
First differencing used to remove the upward trend.

3 Compare

ARIMA and SARIMA candidates compared using AIC/BIC and diagnostics.

- Candidate models included non-seasonal ARIMA and seasonal SARIMA specifications.
- Because the series is seasonally adjusted, a non-seasonal ARIMA model may be sufficient.
- Final selection used the course model-building logic: specification → estimation → diagnostics → forecasting.

Model Fitting and Selection



Selected model

ARIMA(0,1,1) on log turnover

Lowest AIC = -2941.13

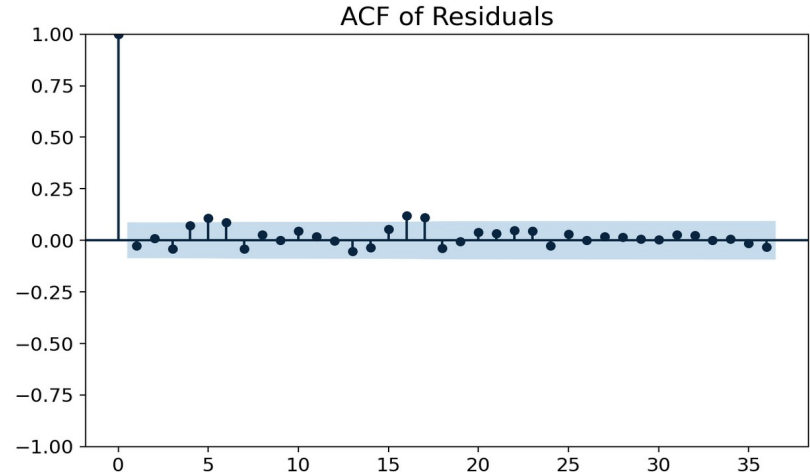
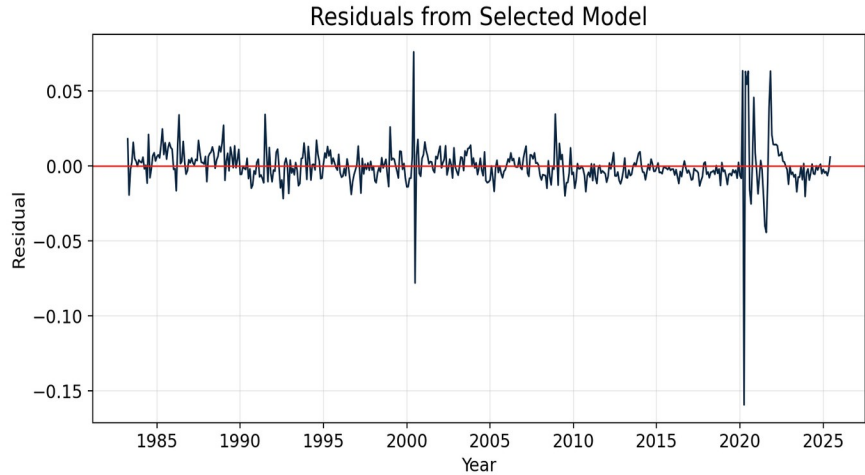
BIC = -2928.39

MA(1) = -0.5099

Drift $\approx$ 0.0046

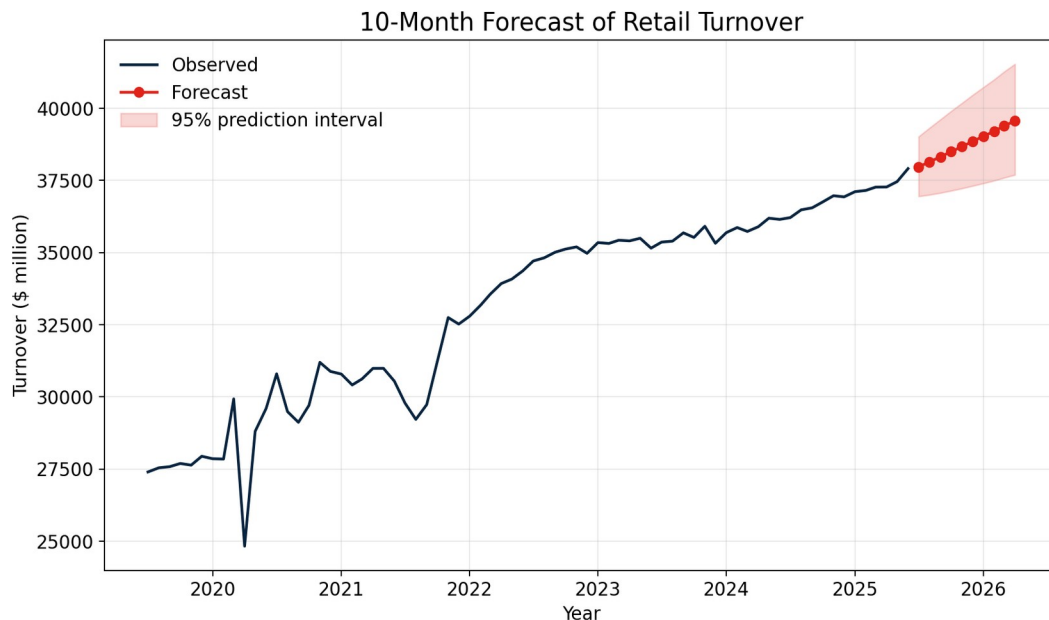
- AIC supported the simpler non-seasonal ARIMA model.
- Seasonal SARIMA terms did not improve the fit for the seasonally adjusted series.

Residual Analysis



- Residuals fluctuate around zero after differencing and MA smoothing.
- Residual ACF shows no major remaining short-lag structure.
- Ljung–Box test at lag 12: $p = 0.18$, so residual autocorrelation is not significant.
- Extreme shocks remain visible, so forecasts should be interpreted with uncertainty.

10-Month Forecast and Conclusion



Main finding

Selected model: ARIMA(0,1,1) on log turnover

Forecasts rise from \$37.97bn in Jul 2025 to \$39.56bn in Apr 2026.

- Retail turnover is forecast to continue increasing gradually.
- Prediction intervals widen over the forecast horizon.
- Sources: ABS Retail Trade, Australia; MATH1318 materials. AI-assisted drafting checked against model outputs.